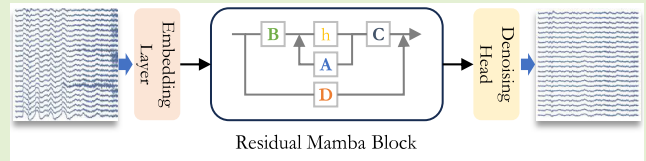


Efficient Ocular Artifacts Removal From EEG Recordings Using State-Space Model

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Abstract—Electroencephalography (EEG) signals are highly susceptible to various artifacts. Eliminating ocular artifacts is crucial for real-world applications, as they predominantly affect prefrontal channels, which are vital for long-term monitoring. Typically, ocular artifacts removal relies on effectively modeling complex relationships within long signal sequences. The mainstream approach currently relies on self-attention mechanisms, but their quadratic computational complexity limits practical applications, especially on mobile devices. To improve efficiency while maintaining effective long-sequence modeling, we introduce an efficient ocular artifacts removal network (EORNet) that leverages the state-space model (SSM) to achieve linear complexity in modeling temporal relationships. EORNet extracts a wide range of EEG features by leveraging the inherent interdependencies within each segment through supervised training, significantly reducing model complexity without compromising representational capabilities. Experimental results demonstrate that EORNet outperforms previous attention-based algorithms, increasing the signal-to-noise ratio by 1.486, reducing floating-point operations (FLOPs) by 36 $\times$, and lowering parameter count by 12 $\times$. Furthermore, EORNet exhibits a 2.57% improvement in motor imagery classification accuracy and a 2.03% enhancement in emotion recognition accuracy compared with raw data, demonstrating its effectiveness in brain-computer interface tasks. This study emphasizes the capability of SSMs in effectively processing time-series biosignals.

Index Terms—Brain-computer interface (BCI), electroencephalogram, electroencephalography (EEG) denoising, ocular artifact removal.



I. INTRODUCTION

ELECTROENCEPHALOGRAPHY (EEG) is a widely utilized technique for measuring neuronal electrical activity and recording brain activity. Its key benefits include noninvasiveness, easy portability, and high temporal resolution [1], making it a favored choice within the domain of brain-computer interfaces (BCIs). EEG plays a pivotal role in neuroscience research and clinical practice, with diverse applications such as seizure prediction [2], emotion recognition [3], motor imagery [4], automatic drowsiness detection [5], sleep stage classification [6], and mental health monitoring [7]. However, EEG signals are highly vulnerable to interference from various types of artifacts. These artifacts

contain electrooculographic (EOG), electromyographic, and electrocardiographic signals coming from eye blinks, muscle movements, and cardiac rhythms, respectively. Among them, EOG artifacts are the most frequently encountered during EEG data acquisition [8]. They are most prominent in prefrontal channels, which are crucial for long-term monitoring [9]. These artifacts significantly degrade EEG quality and compromise diagnostic accuracy. Therefore, it is essential to eliminate ocular artifacts to preserve authentic brain activity during EEG preprocessing.

Various ocular artifact removal techniques have been introduced. One straightforward method involves instructing subjects to minimize unnecessary blinking or eye movements, but maintaining this behavior for a long period is challenging [10]. Another approach is to discard all artifact-contaminated epochs, which risks losing crucial information. To effectively suppress the artifacts, several traditional denoising methods have been proposed, including regression [11], blind source separation (BSS) [13], wavelet transform [14], and hybrid methods that combine multiple techniques [15]. Regression [11] leverages reference signals to calculate contamination coefficients for artifact removal. BSS methods [13] decompose the EEG into components and remove artifact-related ones to eliminate artifacts. Wavelet transform [19] projects the EEG into the wavelet domain, where artifact-related coefficients are selectively eliminated. Hybrid

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methods [15] combine multiple traditional denoising techniques to achieve better artifact suppression ability.

Recently, deep learning-based ocular artifacts removal approaches have emerged as promising methods, as they can automatically eliminate artifacts and have achieved impressive results [16]. A number of approaches have been developed, including autoencoders [17], convolutional neural networks (CNNs) [18], generative adversarial networks [16], recurrent neural networks [27], transformers [28], and hybrid architectures [19], [20]. Specifically, autoencoders have been employed for feature extraction and clean signal reconstruction [17]. CNNs are utilized for learning mappings from contaminated EEG to clean EEG [18]. Transformers enhance global representation learning for artifact removal [28]. GANs are leveraged to improve EEG reconstruction quality [16]. More recent studies have explored hybrid architectures to capture more robust representations, such as combining CNNs with long short-term memory (LSTM) [19] or integrating CNNs and transformers [20].

Previous research has verified the superior performance of transformers in the task of removing EOG artifacts [20]. Despite their effectiveness, transformer-based networks often come with an expensive computational burden, potentially limiting their applications. This is primarily due to the quadratic complexity in extracting attention directly from the temporal dimension of EEG signals. With the increasing demand for ambulatory EEG monitoring, it is essential to design efficient and practical artifact removal networks. Recently, the state-space sequence model [22] has been proposed as a promising framework for efficiently handling long sequences. Based on this model, Mamba [23] employs an efficient hardware-aware algorithm to enhance training efficiency, achieving global receptive fields with linear complexity. Mamba-based models have shown outstanding performance with linear complexity across a range of tasks, including computer vision tasks [24] and biomedical image processing [25]. Inspired by Mamba, we aim to develop a computationally efficient and high-performance EEG denoising network based on the state-space sequence model for practical applications. The key contributions of this article are as follows.

- 1) We introduce an efficient ocular artifacts removal network (EORNet), which leverages residual Mamba blocks (RMBs) as its core module. EORNet effectively captures intricate dependencies within input signals to achieve superior denoising performance.
- 2) We explore the impact of artifact removal on BCI applications and evaluate the effectiveness of EORNet in enhancing motor imagery classification and emotion recognition performance. The results validate its potential for improving the accuracy of BCI tasks.
- 3) Experimental results on public datasets reveal that EORNet outperforms its counterparts across both single-channel and multichannel data while maintaining low computational complexity. This highlights its practical utility for actual applications.

This work is structured as follows. Section II describes the relevant works. In Section III, we introduce a detailed descrip-

tion of EORNet. Section IV provides the evaluation metrics and available datasets. Section V presents the experimental results and related discussions. Finally, we provide a summary of our contributions.

II. RELATED WORKS

A. EEG Denoising Methods

1) *Traditional Artifact Removal Methods*: The traditional denoising methods typically include regression [11], filtering [12], BSS [13], decomposition [14], and hybrid methods [15].

Regression analysis [11] aims to determine mathematical relationships between reference EOG signals and contaminated EEG signals, which is applicable only when reference channels are available. Filtering can be classified into linear filtering and nonlinear adaptive filtering [18]. As in-band artifacts, EOG artifacts are difficult to completely remove by linear filtering, which may result in distortion of EEG signals or information loss. Adaptive filtering approaches are widely used to remove ocular artifacts, which also require additional reference signals. BSS-based approaches [13] have previously been rather popular. Independent component analysis (ICA) first decomposes the EEG signal into mathematically independent components, subsequently removing artifact-related components. However, BSS methods may obtain unreliable results when applied to datasets with few channels.

Several decomposition techniques have been proposed for single-channel and few-channel EEG signals, including wavelet transform and empirical mode decomposition (EMD). In wavelet decomposition [14], EEG signals are projected into the wavelet domain. Subsequently, wavelet coefficients related to artifacts are selectively eliminated while those corresponding to EEG signals are preserved based on a predefined threshold. EMD first decomposes the input EEG into intrinsic mode functions (IMFs). The EEG-related IMFs are utilized to reconstruct denoised signals. In addition, hybrid methods like wavelet-ICA [15] and EMD-ICA [26] have been proposed, exhibiting better abilities in suppressing artifacts while preserving brain activity compared with individual methods alone.

2) *Deep Learning-Based Artifact Removal Methods*: With the preliminary success of deep learning in computer vision and medical image processing, researchers have proposed ocular artifacts removal methods based on deep learning.

In the early stage, denoising networks primarily relied on simple architectures, such as autoencoders, CNNs, and transformers. Yang et al. [17] employed a sparse autoencoder with three hidden layers as feature extractor to eliminate EOG artifacts. Sun et al. [18] captured nonlinear features from input signals using three parallel convolutional branches. Zhang et al. [19] presented four simple networks as basic comparison networks. Sawangjai et al. [16] employed a GAN to enhance the removal of ocular artifacts, where the generator consists of 16 residual blocks. Pu et al. [20] applied transformer for EEG denoising.

As denoising technology advances, more sophisticated and hybridized models are proposed to incorporate advantages of various methods to enhance denoising performance. Gao et al. [15] utilized two CNN branches to capture

multiscale features, followed by two dense layers and one LSTM layer to capture temporal dependencies. Each CNN branch contains five convolutional layers. Chuang et al. [28] introduced IC-U-Net by merging ICA with Unet. Building upon the encoder–decoder architecture of Unet, IC-U-Net first decomposes input signals into brain and nonbrain components using ICA, then extracts multiscale features from these components. Li et al. [29] introduced segmentation into EEG denoising to prevent the distortion of clean EEG segments. Pei et al. [30] proposed DTPNet, integrating encoder–decoder structure with dilated convolutions to capture multiscale contextual information. Cui et al. [31] recovered the EEG signal and artifacts through two branches, subsequently fusing outputs from both branches to obtain more refined reconstructed signals. Yin et al. [21] proposed GCTNet to learn ensemble features from both CNN and transformer branches, with the generator composed of five CNN-transformer blocks. Overall, there is increasing interest in developing various EEG denoising methods.

B. State-Space Models (SSMs)

SSMs serve as foundation frameworks across diverse fields, including control theory and computational neuroscience [22]. Recently, the integration of SSMs with deep learning has become a promising approach for processing long-sequence data.

Earlier, SSMs typically required more memory and encountered vanishing gradients during training, which limited their broader applications. Considering this issue, Gu et al. [22] proposed the structured state-space sequence model (S4), which parameterizes the state transition matrix $\bar{A}$ as a diagonal low-rank matrix, improving computational efficiency and overall performance. In addition, S4 used high-order polynomial projection operators to compute the convolution kernel, enabling more effective long-range dependency modeling while maintaining faster processing speeds. Then Mehta et al. [32] extended S4 by incorporating a gated state-space mechanism, further enhancing training efficiency. Furthermore, Mamba [23] introduced an input-dependent selection mechanism for adaptive information filtering and optimizes the model for hardware efficiency through a parallel algorithm. These advancements significantly improve the practical usability of SSMs in sequential data processing tasks.

The formal definition of SSM in continuous-time representation is given in (1). It maps the continuous input signal $u(t)$ to the hidden state $h(t) \in \mathbb{R}^N$, which is subsequently projected onto the output signal $\tilde{y}(t)$. Here, N denotes the predefined hidden state dimension

$$\begin{aligned} h'(t) &= Ah(t) + Bu(t) \\ \tilde{y}(t) &= Fh(t) + Gu(t) \end{aligned} \quad (1)$$

where $A \in \mathbb{R}^{N \times N}$, $B \in \mathbb{R}^{N \times 1}$, $F \in \mathbb{R}^{1 \times N}$, and $G \in \mathbb{R}^1$ are parameter matrices. $h'(t)$ represents the derivative of $h(t)$. G typically acts as a residual connection and is usually omitted for simplicity.

To process discrete tokens, the continuous formulation above can be discretized with a step size Δ using zero-order

TABLE I
NOTATIONS USED IN THIS ARTICLE

Definition	Notation	
	Single-channel	Multi-channel
Temporal points	$T, t \in \{0, \dots, T-1\}$	$T, t \in \{0, \dots, T-1\}$
Channels	-	$C, c \in \{0, \dots, C-1\}$
Contaminated EEG	$X, x_t \in X$	$X, x_{c,t} \in X$
Clean EOG	$Z, z_t \in Z$	$Z, z_{c,t} \in Z$
Clean EEG	$Y, y_t \in Y$	$Y, y_{c,t} \in Y$
Denoised EEG	$\hat{Y}, \hat{y}_t \in \hat{Y}$	$\hat{Y}, \hat{y}_{c,t} \in \hat{Y}$

hold technique [22], as follows: $\bar{A} = \exp(\Delta A)$, $\bar{B} = (\Delta A)^{-1}(\exp(\Delta A) - I) \cdot \Delta B$, and $\bar{F} = F$. Then given an input sequence $u \in \mathbb{R}^S$, the discrete SSM can be defined as follows:

$$\begin{aligned} h_s &= \bar{A}h_{s-1} + \bar{B}u_s \\ \tilde{y}_s &= \bar{F}h_s \end{aligned} \quad (2)$$

where $\bar{A} \in \mathbb{R}^{N \times N}$, $\bar{B} \in \mathbb{R}^{N \times 1}$, and $\bar{F} \in \mathbb{R}^{1 \times N}$ are discrete parameter matrices.

Assuming the initial hidden state $h_{-1} = 0$, the above formulation can be unrolled as

$$\tilde{y}_s = \sum_{j=0}^s \bar{F} \bar{A}^j \bar{B} u_{s-j}. \quad (3)$$

Then $\tilde{y}$ can be represented as a structured convolution kernel $\bar{K}$ ($\bar{K} \in \mathbb{R}^S$) over the input sequence u , as shown in the following equation:

$$\begin{aligned} \bar{K} &= (\bar{F}\bar{B}, \bar{F}\bar{A}\bar{B}, \dots, \bar{F}\bar{A}^{S-1}\bar{B}) \\ \tilde{y} &= u * \bar{K}. \end{aligned} \quad (4)$$

Therefore, for a 1-D input sequence u , the output of SSM can be expressed as $\tilde{y} = \text{SSM}_{\bar{A}, \bar{B}, \bar{F}}(u)$, which simplifies to $\tilde{y} = \text{SSM}(u)$.

The SSM effectively models long-range dependencies within input sequence, making it well-suited for EEG denoising tasks. In this study, our primary objective is to enhance the efficiency and accuracy of EEG denoising models. To achieve this, we integrate Mamba blocks into the network, leveraging their ability to suppress artifacts while maintaining computational efficiency.

III. METHOD

The pipeline can be described as follows: we initially employ a convolution layer to project the input EEG signals into the feature space and generate tokens by embedding EEG patches. A sequence of RMBs is adopted to further extract the features. These blocks are pivotal components of the proposed network, designed to model and extract inherent long-range temporal dependencies. During this process, the size of feature maps remains consistent, ensuring coherence in signal representation. Subsequently, the features are processed by a denoising head to map the feature map into denoised EEG signals.

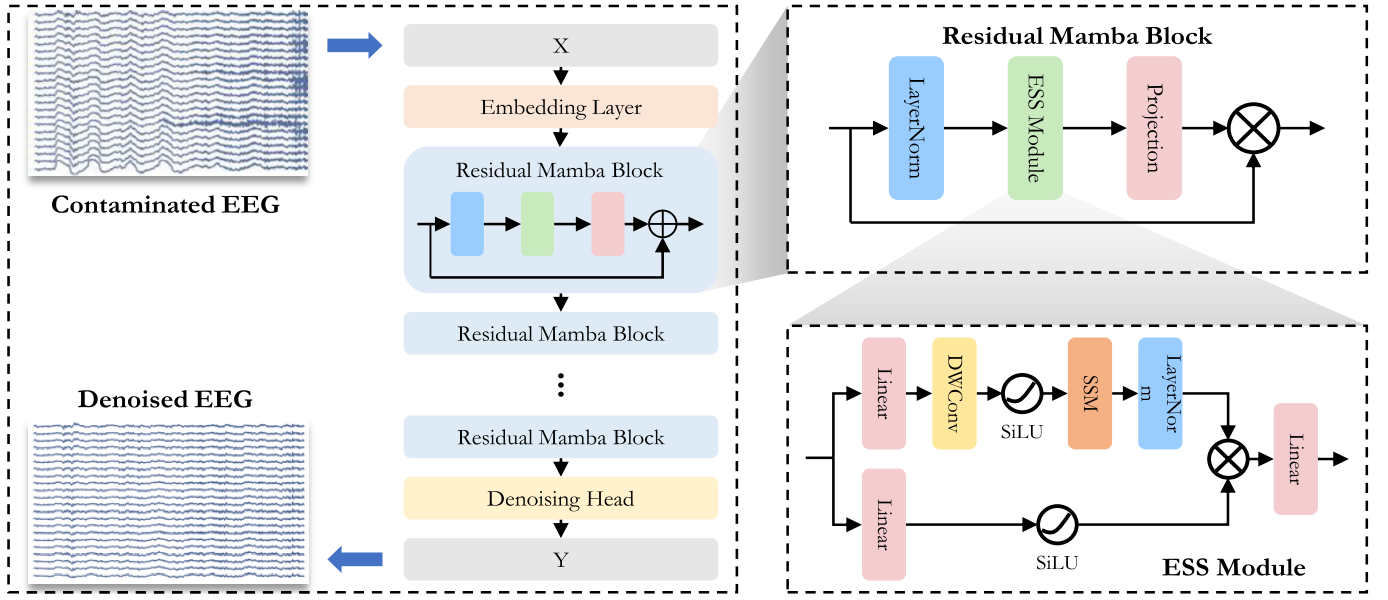


Fig. 1. Network structure of the proposed EORNet. ESS represents the ESS module.

A. Feature Embedding

When given a sample pair $\{(X, Y) | X \in \mathbb{R}^{C \times T}, Y \in \mathbb{R}^{C \times T}\}$, and X represents the contaminated EEG, Y indicates clean EEG, C denotes the number of channels, and T indicates the temporal points. The fundamental mathematical expression for EEG denoising can be defined as

$$X = Y + \Lambda Z \quad (5)$$

where Λ is the mixing matrix. The notation used in this article is as shown in Table I. EORNet first employs a convolution layer for shallow feature extraction, effectively embedding EEG patches and generating tokens. The shallow feature map $F_s \in \mathbb{R}^{D \times T}$ is generated, where D represents a fixed number of filters. The obtained EEG patches are fed into a stack of RMBs, facilitating further processing.

B. Residual Mamba Block

EORNet uses RMB as a basic module to enhance EEG feature extraction. Specifically, RMB utilizes residual connections and linear projection to enhance the extraction of long-range spatial features while maintaining low computational complexity. The shortcut connections can facilitate the recovery of intricate details in the EEG signals without increasing the model parameters, leading to more comprehensive feature extraction.

As depicted in Fig. 1, the RMB begins by receiving the input feature and first applies layer normalization (LayerNorm) to standardize the feature distribution. Then the EEG state-space (ESS) module is employed to extract the spatial dependency features. Subsequently, a linear projection layer converts these extracted features into a deep feature representation. The above process can be expressed as

$$M_i^{\text{out}} = \text{Projection} \left(\text{ESS} \left(\text{LayerNorm} \left(M_i^{\text{in}} \right) \right) \right) \quad (6)$$

where M_i^{in} and M_i^{out} represent the input and output of i -th RMB, respectively. In our experiment, we use four RMBs as feature extractors.

1) **ESS Module:** The ESS module begins with an input feature map, denoted as $P_{\text{in}} \in \mathbb{R}^{D \times T}$, where D is the channel dimension, and T is the temporal dimension. The input feature map is fed into two distinct processing pathways.

In the first branch, the feature map is first expanded using a linear layer to a size of $D \times ET$, where E is a predefined factor, and we set $E = 2$ in our implementation. Subsequently, the feature map is processed using depthwise convolution (DWConv), sigmoid linear unit (SiLU) activation function, SSM, and LayerNorm. The SSM is implemented based on (4), and we set the hidden state dimension to $N = 32$ in our implementation. Both the input and output dimensions of the SSM are consistent, each with a shape of $D \times ET$. In the other branch, the feature map is first expanded to $D \times ET$ using a linear layer, then passed through the SiLU activation function.

The outputs from both branches, denoted as P_1 and P_2 , are then aggregated through a Hadamard product, subsequently fed through a linear layer to produce the output P_{out} . P_{out} and P_{in} have the same feature sizes, i.e. $P_{\text{out}} \in \mathbb{R}^{D \times T}$. The entire process is described as follows:

$$\begin{aligned} P_1 &= \text{LayerNorm}(\text{SSM}(\text{SiLU}(\text{DWConv}(\text{Linear}(P_{\text{in}})))))) \\ P_2 &= \text{SiLU}(\text{Linear}(P_{\text{in}})) \\ P_{\text{out}} &= \text{Linear}(P_1 \odot P_2) \end{aligned} \quad (7)$$

where $\odot$ is the Hadamard product.

RMB plays a crucial role in capturing and modeling the intricate long-range dependencies inherent in input EEG features. By effectively using the capabilities of Mamba, the proposed approach can better understand and represent the complex interrelations in EEG data, thus obtaining accurate and efficient denoising performance.

TABLE II
DATASETS USED IN THIS WORK

Datasets	Subjects	Sampling Rates	Channels	Usages
EEGDenoiseNet [27]	90	256 Hz	1	Assess the single-channel denoising performance
Semi-simulated EEG/EOG dataset [33]	27	200 Hz	19	Assess the multi-channel denoising performance
BCI 2b [34]	9	250 Hz	3	Assess the impact of denoising on motor imagery task
SEED-VII [35]	20	250 Hz	62	Assess the cross-dataset denoising performance on emotion recognition task
DEAP [36]	32	250 Hz	32	Train the denoising model for testing on the SEED-VII dataset

C. Denoising Head

The denoising head plays a role in transforming the output feature map into denoised EEG signals. The entire conversion process can be expressed as

$$\hat{Y} = g_h(M_L^{\text{out}}) \quad (8)$$

where M_L^{out} represents the output feature map after L RBMs. The mapping function of the denoising head is represented by g_h . In our approach, we employ two convolutional layers without activation functions as the denoising head.

The training loss is defined as mean squared error (mse) loss. After being processed by EORNet, the formula of mse loss can be mathematically expressed as

$$L_{\text{mse}} = \mathbb{E}_{X, Y \sim P(X, Y)} [\|Y - \hat{Y}\|_F^2]. \quad (9)$$

The model is trained using iterative optimization, employing supervised learning to minimize the discrepancy between the estimated and clean EEG signals.

IV. EXPERIMENTS

A. Datasets

Five public datasets are employed to evaluate the EORNet's performance in ocular artifacts removal. EEGDenoiseNet [27] and the semi-simulated EEG/EOG dataset [33] are used to assess single-channel and multichannel EEG denoising performance, respectively. The BCI Competition IV Dataset 2b [34] is utilized to evaluate the impact of denoising on motor imagery task. Shanghai Jiao Tong University Emotion EEG Dataset VII (SEED-VII) [35] and a database for emotion analysis using physiological signals (DEAP) [36] are employed to assess EORNet's cross-dataset denoising performance on the emotion recognition task. These datasets provide a comprehensive evaluation of EORNet's performance. Detailed descriptions are provided in Table II.

1) *EEGDenoiseNet*: It is a single-channel EEG denoising dataset, which comprises 4514 clean EEG and 3400 clean EOG segments. Each segment lasts two seconds and is sampled at 250 Hz. The contaminated data is generated by linearly combining clean EEG and clean EOG

$$x = y + \lambda z \quad (10)$$

where x , y , and z denote the contaminated EEG, clean EEG, and artifact signals, respectively. The parameter λ represents

the contamination factor for controlling contamination levels. Signal-to-noise ratio (SNR) is determined according to

$$\text{SNR (dB)} = 10 \log \frac{P_Y}{P_{X-Y}} = 10 \log \frac{\sum_{t=0}^{T-1} y_t^2}{\sum_{t=0}^{T-1} (\lambda \cdot z_t)^2} \quad (11)$$

where $P(\cdot)$ is the power. Following prior studies, the SNR values are adjusted to be between -5 and 5 dB, increasing at intervals of 1 dB [18], [20].

2) *Semi-Simulated EEG/EOG Dataset*: It directly provides clean EEG signals and contaminated EEG signals polluted by EOG artifacts. It encompasses data from 19 channels across 54 recordings from 27 subjects. All recordings last for a duration of 30 s (200 Hz). During preprocessing, EOG signals are filtered with a 0.5- to 5-Hz bandpass filter. EEG signals are filtered with a 0.5- to 40-Hz bandpass filter and a 50-Hz notch filter. All EEG signals are carefully inspected by the provider to ensure the absence of substantial contamination. Contaminated EEG signals are obtained through linear combination of clean EEG, horizontal EOG, and vertical EOG signals.

3) *BCI Competition IV Dataset 2b*: It is a motor imagery task dataset, in which EEG signals are classified into two classes: left-hand or right-hand motor imagery. The data were collected from nine individuals, each contributing both training and test sets. All data is sampled at 250 Hz from three channels (C3, Cz, and C4). The preprocessing includes a 0.5–100-Hz bandpass filter and a 50-Hz notch filter. Each recording segment lasts for 9 s and is divided into distinct phases, including fixation, cue appearance, imagery period, and pause. We extract the EEG signals during the imagery period for classification, aligning with the method in [16]. In addition, this dataset provides EOG recordings from three monopolar electrodes.

4) *SEED-VII*: The SEED-VII dataset is collected by Shanghai Jiao Tong University and comprises EEG recordings from 20 subjects across seven emotion categories. The EEG data is recorded at 1000 Hz using a 62-channel device with international 10–20 system. During the experiment, each participant watches 80 video clips, with each clip lasting between 2 and 5 min, designed to elicit specific emotional responses. Following the preprocessing method described in [35], a band-pass filter (0.1–70 Hz) was applied, followed by a 50-Hz notch filter to eliminate powerline interference. All EEG signals were downsampled to 250 Hz.

5) *DEAP*: It is a multimodal dataset for emotion analysis, containing EEG, EOG, and EMG data. It contains EEG recordings gathered from 32 individuals, with each participant completing 40 experimental trials. The EEG signals are recorded at 32 channels at a 512-Hz sampling rate. The preprocessing includes a 0.1–70-Hz bandpass filter, a 50-Hz notch filter, and downsampling to 250 Hz to maintain consistency with the SEED-VII dataset.

B. Performance Metrics

We assess EORNet on three aspects: the quality of ocular artifact removal, the effectiveness in BCI classification tasks, and the efficiency in eliminating ocular artifacts.

1) *Quality of Ocular Artifact Removal*: We first verify the quality of the generated denoised EEG on two semi-simulated datasets. The relative root mean square error (RRMSE) is computed to measure the difference between the denoised and clean signals, with lower RRMSE values indicating better denoising performance. Subsequently, we compute the correlation coefficient (CC) and SNR for these signals, where higher CC and SNR values reflect better denoising effectiveness.

RRMSE is expressed as follows:

$$\text{RRMSE} = \frac{\sqrt{\sum_{c=0}^{C-1} \sum_{t=0}^{T-1} (\hat{y}_{c,t} - y_{c,t})^2}}{\sqrt{\sum_{c=0}^{C-1} \sum_{t=0}^{T-1} (y_{c,t})^2}}. \quad (12)$$

CC is the Pearson CC, defined as

$$\text{CC} = \frac{\text{cov}(\hat{Y}, Y)}{\sigma_{\hat{Y}} \sigma_Y} \quad (13)$$

where σ represents standard deviation and cov denotes covariance. CC reflects the similarity between $\hat{Y}$ and Y , with values closer to 1 indicating better denoising performance.

SNR is calculated as

$$\text{SNR (dB)} = 10 \log_{10} \frac{P_Y}{P_{Y-\hat{Y}}}. \quad (14)$$

The above metrics can quantitatively validate the learning capacity of the model in EOG artifact removal task.

2) *Effectiveness in BCI Classification Tasks*: The genuine EEG data lack labels, making it impossible to calculate the aforementioned criteria. Therefore, we evaluate the proposed network's performance on the BCI classification task. Evaluation metrics for classification tasks include accuracy and the $F1$ -score

$$\begin{aligned} \text{Accuracy} &= \frac{\text{TP} + \text{TN}}{\text{TP} + \text{FP} + \text{TN} + \text{FN}} \\ \text{Recall} &= \frac{\text{TP}}{\text{TP} + \text{FN}} \\ \text{Precision} &= \frac{\text{TP}}{\text{TP} + \text{FP}} \\ F1 &= \frac{2 \times \text{Recall} \times \text{Precision}}{\text{Precision} + \text{Recall}} \end{aligned} \quad (15)$$

where TP, FP, TN, and FN are true positive, false positive, true negative, and false negative, respectively.

Furthermore, we conduct an analysis of the power spectral density (PSD) of the denoised data obtained by

TABLE III

EXPERIMENTAL RESULTS (MEAN $\pm$ STD) OF THE EEGDENOISENET ARE SHOWN. **BOLD** DENOTES BEST RESULTS. UNDERLINE INDICATES SECOND-BEST RESULTS

Methods	RRMSE	CC	SNR
EMD-ICA	0.780 $\pm$ 0.006	0.792 $\pm$ 0.003	0.769 $\pm$ 0.075
SimpleCNN	0.273 $\pm$ 0.008	0.947 $\pm$ 0.004	12.717 $\pm$ 0.282
ResCNN	0.272 $\pm$ 0.034	0.945 $\pm$ 0.004	12.726 $\pm$ 0.270
NovelCNN	0.292 $\pm$ 0.010	0.924 $\pm$ 0.007	13.234 $\pm$ 0.287
DuoCL	0.284 $\pm$ 0.015	0.935 $\pm$ 0.006	12.830 $\pm$ 0.602
GCTNet	0.226 $\pm$ 0.009	0.959 $\pm$ 0.004	15.013 $\pm$ 0.303
Ours	0.198 $\pm$ 0.008	0.966 $\pm$ 0.003	16.499 $\pm$ 0.326

TABLE IV

PERFORMANCE (MEAN $\pm$ STD) FOR THE SEMI-SIMULATED EEG/EOG DATASET IS PRESENTED BELOW. **BOLD** DENOTES BEST RESULTS. UNDERLINE INDICATES SECOND-BEST RESULTS

Methods	RRMSE	CC	SNR
ICA	0.390 $\pm$ 0.118	0.909 $\pm$ 0.041	10.532 $\pm$ 2.538
SimpleCNN	0.287 $\pm$ 0.042	0.941 $\pm$ 0.020	11.687 $\pm$ 1.263
ResCNN	0.272 $\pm$ 0.034	0.948 $\pm$ 0.016	12.086 $\pm$ 1.102
NovelCNN	0.296 $\pm$ 0.030	0.943 $\pm$ 0.014	11.026 $\pm$ 0.874
DuoCL	0.294 $\pm$ 0.045	0.942 $\pm$ 0.019	11.350 $\pm$ 1.408
GCTNet	0.264 $\pm$ 0.032	0.952 $\pm$ 0.012	12.323 $\pm$ 1.097
EEGNet	0.206 $\pm$ 0.044	0.963 $\pm$ 0.018	14.667 $\pm$ 1.456
Ours	0.153 $\pm$ 0.018	0.978 $\pm$ 0.007	17.195 $\pm$ 1.020

various networks. PSD is a widely used assessment metric for actual data. The PSD of the processed EEG should decrease within 0.5–12 Hz, while those in other frequency ranges should remain close to the original EEG signals.

3) *Efficiency of Ocular Artifacts Removal*: The aforementioned metrics are utilized to evaluate denoising performance on semi-simulated and real-world datasets. As long-term monitoring becomes more prevalent, the efficiency of EOG artifact removal is an important consideration. Therefore, we evaluate computational complexity using floating-point operations (FLOPs) and inference time on identical hardware configurations.

C. Implementation Details

For the experiments on the first three datasets, the batch size is 32, the number of epochs is set to 50, and the Adam optimizer is utilized with $\beta_1 = 0.9$ and $\beta_2 = 0.99$. The best model is chosen based on its superior denoising performance on the validation set. To reduce the impact of randomness, the quantitative indicators of semi-simulated data are calculated using 10-fold cross-validation method.

1) *For the Experiment on EEGDenoiseNet*: The learning rate is $1e^{-4}$. For each fold, all segments are divided into training and testing subsets according to 9:1. Subsequently, the validation set is created by randomly selecting 10% of data from training set.

2) *For the Experiment on Semi-Simulated EEG/EOG Dataset*: The learning rate is $5e^{-4}$. During training, we randomly select 100 initial points from each recording and extract windows of length 500 to augment training data. The

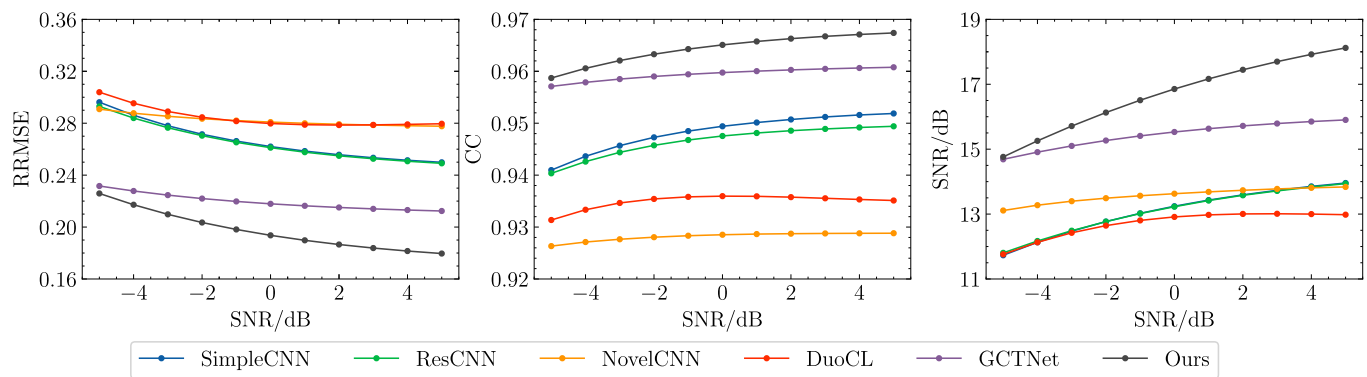


Fig. 2. Comparison of EOG artifact removal results from different networks across various SNRs.

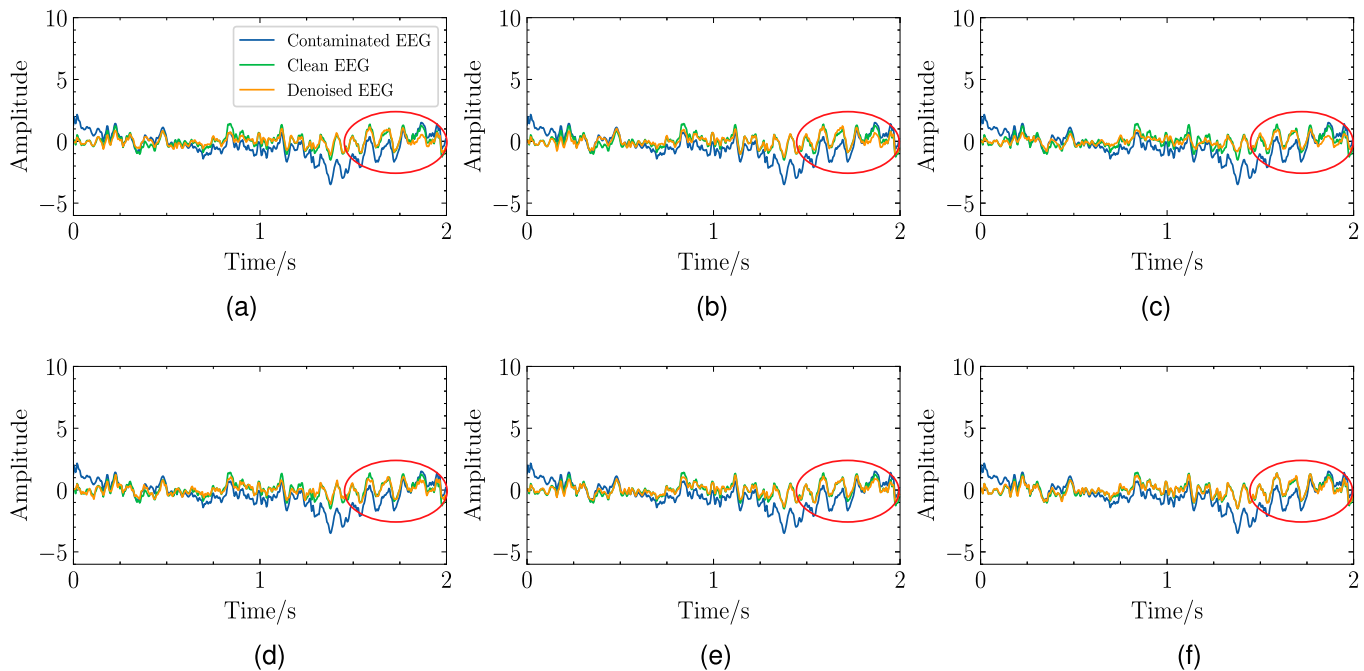


Fig. 3. Comparison of visualization results for EOG artifact removal across different networks. (a) SimpleCNN. (b) ResCNN. (c) NovelCNN. (d) DuoCL. (e) GCTNet. (f) Ours.

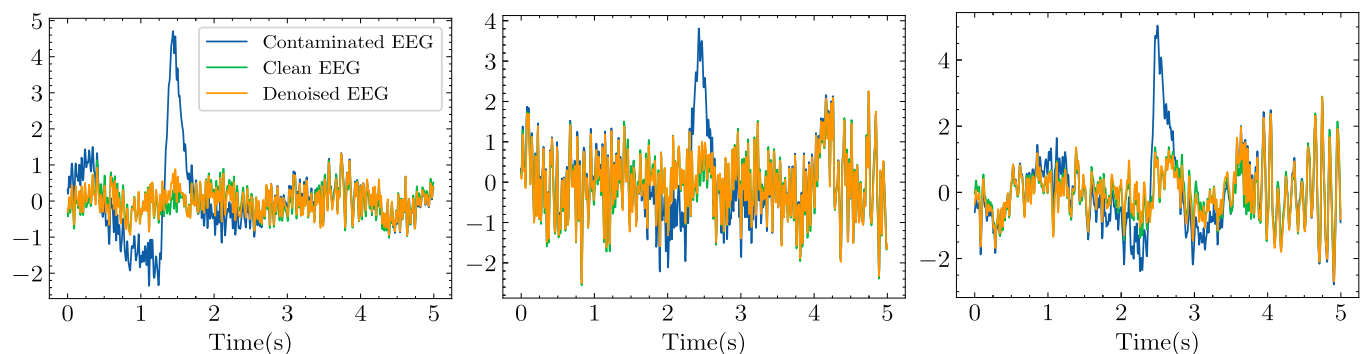


Fig. 4. Visual results on the semi-simulated EEG/EOG dataset.

54 recordings are split into training and test sets according to 9:1, with 10% of the training data randomly selected as the validation set. We train the denoising model as a 19-channel model on this dataset.

3) For the Experiment on BCI 2b: We further evaluate the performance of EORNet on the BCI classification task. Specifically, we aim to improve classification performance by mitigating ocular artifacts. To validate our approach,

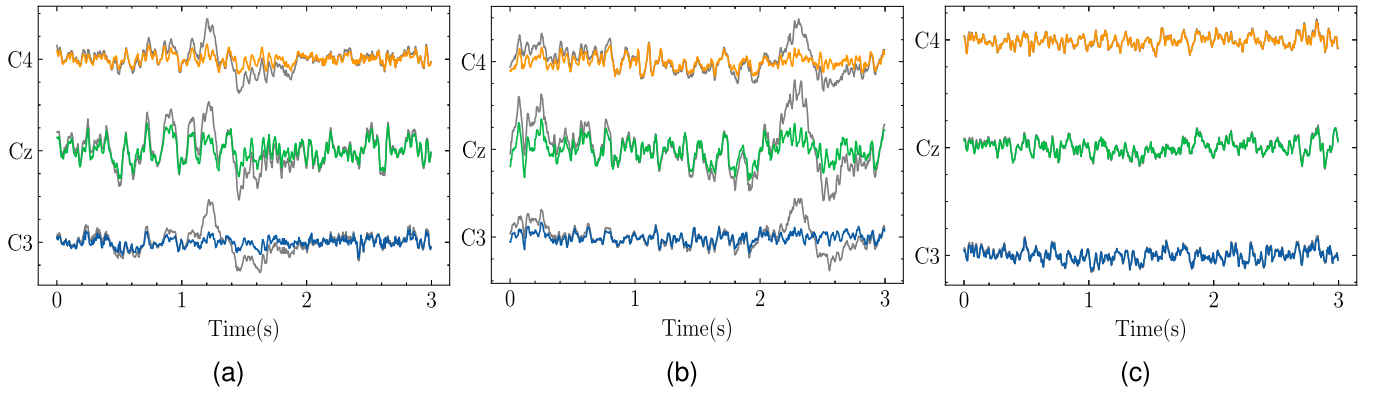


Fig. 5. Visual results on BCI 2b dataset. Raw EEG data (gray) and reconstructed EEG signals for various channels are shown. (a) Heavy contamination. (b) Heavy contamination. (c) Minimal contamination.

TABLE V

CLASSIFICATION RESULTS (MEAN $\pm$ STD) ON BCI 2B DATASET. **BOLD** DENOTES BEST RESULTS. UNDERLINE INDICATES SECOND-BEST RESULTS. *INDICATES A SIGNIFICANT DIFFERENCE ($P < 0.05$) BETWEEN EEG DENOISED BY OUR METHOD AND OTHERS

Artifact Removal Networks	Raw	ResCNN	GCTNet	EEGNet	EORNet
Accuracy (%)	73.01 $\pm$ 11.39*	74.08 $\pm$ 11.53	74.69 $\pm$ 13.90	<u>74.93 $\pm$ 13.24</u>	75.58 $\pm$ 12.50 ($\uparrow$ 2.57)
F1-score (%)	72.41 $\pm$ 12.22*	73.56 $\pm$ 11.70	74.04 $\pm$ 14.34	<u>74.35 $\pm$ 13.60</u>	75.10 $\pm$ 12.76 ($\uparrow$ 2.69)

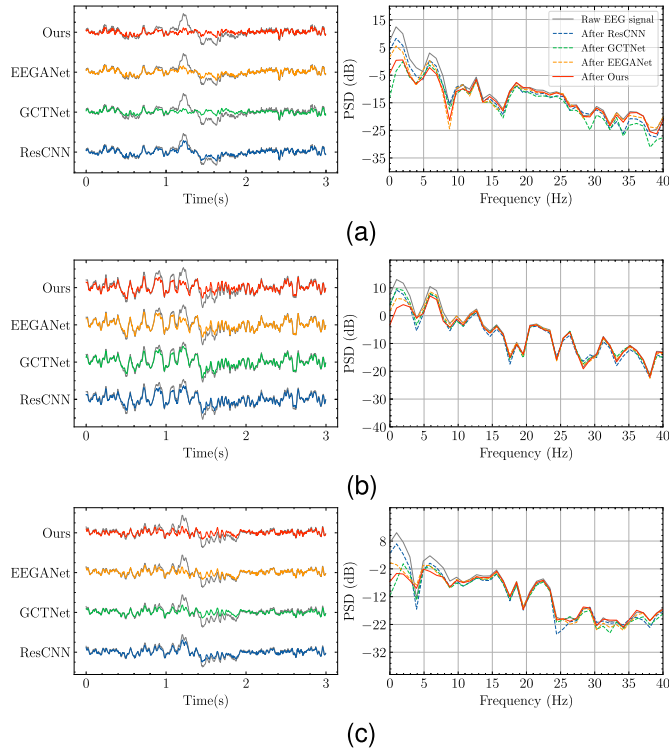


Fig. 6. Typical example [corresponding to Fig. 5(a)] of denoised EEG waveform and PSD after ocular artifact removal. Both the raw EEG signals (depicted in gray) and the reconstructed EEG signals processed by various networks are presented. (a) EOG (C3). (b) EOG (Cz). (c) EOG (C4).

we employ the BCI Competition IV 2b dataset for evaluation. Due to the absence of denoising labels in the raw data, we adopt the regression method introduced by Schlögl et al. [37], which efficiently generates denoised EEG signals using reference channels. This method is also recommended by the instructions of BCI competition IV.

Initially, we use regression to produce denoised signals for the training set. Subsequently, we train the denoising model on training set. The trained model is then applied to generate denoised data for training and test sets. To evaluate the effectiveness of denoising on downstream tasks, we conduct a comparative analysis of the classification performance between raw and denoised data. The learning rate of denoising model is $5e^{-4}$. Since this dataset contains only three EEG channels, EORNet is trained as a three-channel model accordingly. We employ the filter bank common spatial pattern and support vector machine as the classification method. For further implementation details on the classification method, please refer to [16].

4) *For the Experiment on SEED-VII:* We evaluate cross-dataset denoising performance by training the model on DEAP and testing on SEED-VII. Similarly, we aim to enhance classification performance by reducing ocular artifacts. Given that the DEAP dataset comprises 32-channel EEG recordings, whereas the SEED-VII dataset contains 62-channel EEG recordings, we set the input and output channels in EORNet to 1 for compatibility. This allows the model trained on DEAP to be directly evaluated on EEG recordings from the SEED-VII dataset in a channel-by-channel manner. For the denoising task, clean EEG signals are manually selected to create pairs of clean and contaminated EEG segments. Each segment is trimmed to a 10-s duration. This process generates a total of 337408 segments of size 1×2500 . The data is divided into training and testing sets at a ratio of 90% to 10%. During training, ten random initial points are generated for each segment in every epoch. The model input size is 1×500 . The Adam optimizer is employed with a learning rate of $1e^{-4}$, a batch size of 384, and 20 training epochs.

Subsequently, the trained denoising model is applied to SEED-VII to generate denoised data. Regarding the

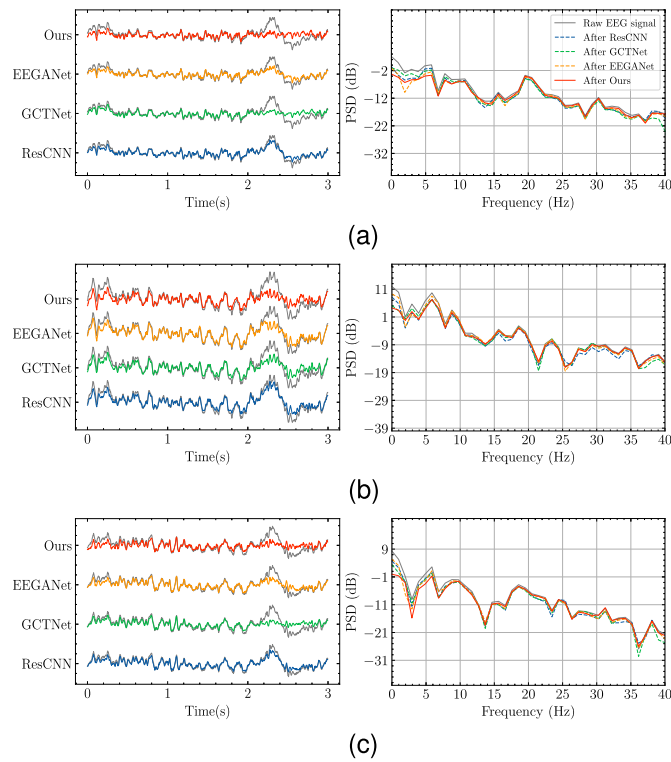


Fig. 7. Typical example [corresponding to Fig. 5(b)] of denoised EEG waveform and PSD after ocular artifact removal. Both the raw EEG signals (depicted in gray) and the reconstructed EEG signals processed by various networks are presented. (a) EOG (C3). (b) EOG (Cz). (c) EOG (C4).

classification method, we employ Tsception [38], a high-performance classification network for emotion recognition tasks. The input EEG data for Tsception is 1 s. The Adam optimizer is used for the classification network, with a learning rate of $1e^{-4}$, a batch size of 64, and 100 training epochs. Each subject in the SEED-VII dataset consists of 80 sessions. To reduce the impact of randomness, fourfold cross-validation is applied, with each fold containing 20 sessions. Two folds are used for training, one for validation, and one for testing. The final experimental results are reported as the average across 20 subjects using fourfold cross-validation.

D. Platform

All experiments are implemented in PyTorch 1.13.1, Python 3.9, and CUDA 11.7.

V. RESULTS AND DISCUSSION

In this section, we comprehensively assess EORNet's performance by evaluating the denoising performance, effectiveness in downstream tasks, cross-dataset denoising performance, and inference efficiency.

A. Performance Comparison on EEGDenoiseNet Dataset

To evaluate the effectiveness of EORNet, we benchmark it against several competitive single-channel denoising methods, specifically: EMD-ICA, SimpleCNN, ResCNN, NovelCNN,

DuoCL, and GCTNet. EMD-ICA is implemented according to [39]. It's noteworthy that recent studies employ network search techniques to determine the optimal number of modules and network layers for different datasets. To ensure a fair comparison, our evaluation focuses solely on state-of-the-art networks with fixed structures, excluding dynamically optimized networks. We quantify the performance according to RRMSE, CC, and SNR.

As shown in Table III, EMD-ICA has the poorest denoising performance, achieving only 0.780, 0.792, and 0.769 in RRMSE, CC, and SNR, respectively. EORNet achieves the lowest RRMSE, highest CC, and superior SNR. For example, in terms of SNR, SimpleCNN, ResCNN, NovelCNN, DuoCL, and GCTNet achieve 12.717, 12.726, 13.234, 12.830, and 15.013 dB, respectively. In comparison, EORNet achieves 16.499 dB, demonstrating its effectiveness in eliminating artifacts and preserving brain information. When compared with GCTNet, EORNet shows an average decrease in RRMSE of over 12%, an average increase in CC exceeding 0.7%, and an average SNR improvement of more than 9%.

We subsequently investigate the variation in denoising performance relative to changes in input SNRs. In Fig. 2, we observe that various deep learning-based artifact methods demonstrate similar performance fluctuations under diverse SNR levels. EORNet consistently achieves the lowest RRMSE, highest SNR, and largest CC values across different input SNR levels, showcasing superior denoising performance compared with other networks. This superiority is particularly evident at high input SNR levels. Such remarkable performance may be attributed to EORNet's unique design, which emphasizes the extraction of global features through the residual Mamba module while avoiding pooling layers to preserve original information.

Fig. 3 provides an example of EOG artifact removal results processed by various networks. The contaminated EEG data is represented by the blue lines, while the clean EEG and denoised signals are depicted by the green and yellow lines, respectively. As shown in Fig. 3, a peak amplitude offset is observed in the results from SimpleCNN, ResCNN, and NovelCNN. When comparing the waveforms generated by DuoCL, GCTNet, and EORNet, it is evident that EORNet demonstrates superior denoising performance due to its lower waveform offset.

B. Performance Comparison on Semi-Simulated EEG/EOG Dataset

In addition to the previously mentioned methods, we include ICA and EEGANet as a comparison method. ICA is a widely used multichannel EEG method. In our implementation, we use ICA for decomposition and employ ICLabel [40] to automatically remove artifact-related components. EEGANet is designed for eliminating multichannel EOG artifacts from EEG recordings.

Table IV presents the statistical results from 10-fold cross-validation. Among compared networks, ICA-ICLabel has the poorest denoising performance compared with other methods, achieving only 0.390 for RRMSE, 0.909 for CC, and 10.532 dB for SNR. EEGANet outperforms other networks in

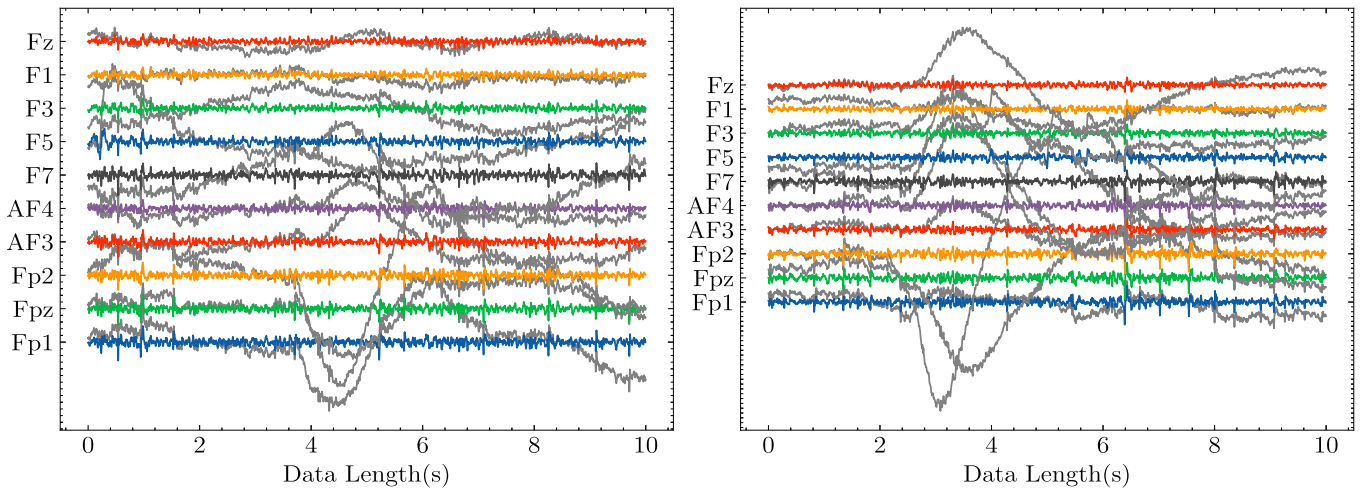


Fig. 8. Visual results on SEED-VII dataset. Raw data (gray) and reconstructed signals (color) across various channels are shown.

denoising. The RRMSE, CC, and SNR values for EEGANet are 0.206, 0.963, and 14.667, respectively, while EORNet achieves 0.153, 0.978, and 17.195 in these metrics, demonstrating a substantial improvement. Specifically, RRMSE decreases by 25.7%, CC improves by 1.56%, and SNR increases by 17.24%, highlighting the effectiveness of EORNet.

The visual results depicted in Fig. 4 illustrate the effectiveness of the denoising process. The generated EEG signals demonstrate consistency with the ground truth, affirming the trained model's capability in eliminating EOG artifacts while preserving the inherent nonlinearity of EEG signals. Moreover, we find that EEGANet and EORNet exhibit significant superiority over other networks. This superiority can be attributed to their utilization of complete multichannel EEG data as input for denoising and incorporation of global information during feature extraction. In contrast, other single-channel denoising network address multichannel signals channel by channel, which may result in information loss.

C. Performance Comparison in BCI Application

Eliminating artifacts is critical in downstream tasks. In this section, we investigate the influence of EEG denoising on motor imagery tasks. Based on the experimental results of semi-simulated dataset, we select three top-performing networks for comparison: ResCNN, GCTNet, and EEGANet.

As shown in Table V, the raw data achieves accuracy and $F1$ -score values of 73.01% and 72.41%, respectively. When compared with raw data, the denoised signals produced by ResCNN, GCTNet, EEGANet, and EORNet achieve accuracy of 74.08%, 74.69%, 74.93%, and 75.58%, respectively. For $F1$, the denoised signals processed by these networks achieve scores of 73.56%, 74.04%, 74.35%, and 75.10%, respectively. These results indicate that denoised EEG signals from various networks exhibit improved classification performance. This experimental result corroborates with [16]. Among them, EORNet-processed signals demonstrate superior precision and $F1$ metrics. We use the Wilcoxon signed-rank test to conduct statistical analysis. EORNet demonstrates a

significant improvement, achieving a 2.57% ($p < 0.05$) increase and a 2.69% ($p < 0.05$) improvement in $F1$ -score compared with raw data inputs. The above results demonstrate the effectiveness of EORNet in improving signal quality and reliability for EEG analyses.

In Fig. 5, we present temporal waveforms processed by EORNet and raw temporal waveforms from the test set. Raw temporal waveforms are plotted in gray, while denoised signals are denoted in blue (C3), green (Cz), and yellow (C4). Fig. 5(a) and (b) shows heavy contamination. We observe that most of the ocular artifacts are successfully suppressed. In Fig. 5(c), minimal contamination is evident. We find that EORNet has nearly no change, indicating EORNet's strong ability to retain the original information for clean data. Furthermore, we depict the waveforms and PSD of raw and denoised EEG signals processed by various networks. Fig. 6 demonstrates that ocular artifacts occurring at 1 and 2 s are effectively removed. Similarly, Fig. 7 shows that the ocular artifacts that appear at 2 and 3 s are successfully suppressed by EORNet. The PSD of EORNet closely aligns with the original signals in the high-frequency band, indicating its potential for signal retention. In addition, it exhibits a greater reduction in the 0.5–12-Hz range, suggesting its powerful denoising capability.

D. Cross-Dataset Denoising Performance in BCI Applications

Besides the aforementioned motor imagery task, we further evaluate cross-dataset denoising performance in the emotion task. We train the model on DEAP dataset with clean and contaminated data, and test its performance on SEED-VII. For comparison, we select the three best-performing single-channel networks based on experimental results on EEGDenoiseNet.

The raw EEG signal contains 62 channels. Fig. 8 displays the visual outcomes for the first ten channels, which are located in the prefrontal region and are highly susceptible to ocular artifact contamination. The contaminated signals are shown in gray, while reconstructed signals are represented

TABLE VI

CLASSIFICATION RESULTS (MEAN $\pm$ STD) ON SEED-VII DATASET. **BOLD** DENOTES BEST RESULTS. UNDERLINE INDICATES SECOND-BEST RESULTS. * INDICATES A SIGNIFICANT DIFFERENCE ($P < 0.05$) BETWEEN EEG DENOISED BY OUR METHOD AND OTHERS

Artifact Removal Networks	Raw	SimpleCNN	ResCNN	GCTNet	EORNet
Accuracy (%)	33.91 $\pm$ 6.97*	35.45 $\pm$ 7.20	35.55 $\pm$ 7.25	<u>35.59 $\pm$ 6.91</u>	35.94 $\pm$ 7.12(↑ 2.03)
F1-score (%)	32.36 $\pm$ 7.31*	33.89 $\pm$ 7.44	33.86 $\pm$ 7.45	<u>33.98 $\pm$ 6.96</u>	34.39 $\pm$ 7.21(↑ 2.03)

TABLE VII

COMPARISON OF FLOPS AND PARAMETER NUMBERS. **BOLD** DENOTES BEST RESULTS. UNDERLINE INDICATES SECOND-BEST RESULTS

Artifact Removal Networks	SimpleCNN	ResCNN	GCTNet	EEGNet	EORNet
FLOPs (G)	0.484	0.605	5.834	<u>0.174</u>	0.164
Params (M)	10.278	5.187	28.058	0.434	<u>2.267</u>
Time (ms)	3.90	8.18	33.87	25.92	<u>6.56</u>

in color. The results demonstrate that EORNet effectively suppresses ocular artifacts, preserving the original EEG signals. For example, ocular artifacts in channels Fp1, Fp2, and Fz between 3 and 7 s are largely eliminated. For the second recording, the ocular artifacts in all ten channels are reduced. We further assess the classification performance of the denoised EEG signals by contrasting the accuracy of the raw EEG data with those processed by several denoising techniques. Table VI shows the classification results for the seven-class emotion recognition task.

Among the four denoising methods, EORNet achieves the highest accuracy and $F1$ -score in cross-dataset denoising, improving accuracy from 33.91% to 35.94% and $F1$ -score from 32.36% to 34.39%. To evaluate the statistical significance of these improvements, we conduct the Wilcoxon signed-rank test. The results indicate that EORNet significantly enhances classification performance, achieving a 2.03% ($p < 0.05$) improvement in both accuracy and $F1$ -score compared with raw EEG. Moreover, all denoising methods exhibit statistically significant improvements over raw EEG, suggesting their effectiveness in mitigating ocular artifacts. This improvement is particularly notable in the SEED-VII dataset, likely due to the high intensity of artifacts contamination (as shown in Fig. 8), which makes artifact removal more beneficial for classification performance. These results demonstrate that EORNet can effectively address the cross-dataset denoising task.

E. Comparison of Efficiency

We then analyze the efficiency of the proposed model by comparing its computational complexity on semi-simulated EEG/EOG dataset. We select the four top-performing denoising networks from previous experiments for comparative analysis. Detailed results for an input length of 2 s are presented in Table VII. The FLOPs for SimpleCNN, ResCNN, GCTNet, EEGNet, and EORNet are 0.484, 0.605, 5.834, 0.174, and 0.164, respectively. EORNet shows the lowest FLOPs, indicating low latency and high computational efficiency. EORNet exhibits a slightly higher parameter count compared with EEGNet, which may lead to a marginal increase in memory usage. However, it remains lower than

other denoising networks, resulting in minimal memory demand during inference. The inference time for SimpleCNN, ResCNN, GCTNet, EEGNet, and EORNet is 3.90, 8.18, 33.87, 25.92, and 6.56 ms, respectively. Although EORNet is slightly slower than SimpleCNN, it outperforms all other networks in speed. Specifically, EORNet processes a 19-channel, 2-s segment in just 6.56 ms. The processing time is substantially less than the input length. These results highlight EORNet's high efficiency and excellent potential in real-world applications.

F. Limitations and Future Work

Although we have achieved satisfactory performance in precision and speed, several aspects still require further improvement. First, cross-dataset distribution differences pose a challenge. Our studies have confirmed that deep learning networks exhibit a degree of generalization across datasets. However, differences in data distribution among various datasets can degrade denoising effectiveness. Therefore, minimizing these distribution differences and improving denoising performance remain critical challenges in ongoing research. Second, EEG is widely utilized in various BCI tasks, including epilepsy monitoring, sleep detection, and stress monitoring. We have verified the effectiveness of EEG denoising in motor imagery and emotion recognition tasks. Moving forward, we aim to apply EEG denoising to a broader range of downstream tasks to enhance its practical applicability. Third, EEG signals are often contaminated by a variety of artifacts, particularly EMG artifacts, which degrade signal quality and affect the outcomes of EEG experiments. The proposed network is specifically designed to process EOG artifacts. In future work, we might extend our research to develop a more comprehensive denoising network capable of suppressing multiple types of artifacts, including EMG artifacts.

VI. CONCLUSION

This article introduces an efficient EOG artifact removal network that utilizes a RMB as its core module. It can effectively eliminate EOG artifacts and achieve superior denoising performance on single-channel and multichannel datasets. Moreover,

EORNet achieves enhanced classification accuracy in both motor imagery and emotion tasks, indicating its practical utility in real-world applications. When conducting an efficiency analysis of various networks, we observe that EORNet obtains superior accuracy with low computational complexity. This work is an attempt to integrate SSM with EEG denoising, which successfully achieves excellent performance and significantly enhances the computational efficiency. Mamba is a promising network for EEG data analysis with advantages of long-range information extraction ability and high computational efficiency.

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